

Fig. 6.27. *Two Networks in Tandem*

$$\mathbf{A} = \mathbf{A_1A_2} + \mathbf{B_1C_2}$$
$$\mathbf{B} = \mathbf{A_1B_2} + \mathbf{B_1D_2}$$
$$\mathbf{C} = \mathbf{A_2C_1} + \mathbf{C_2D_1}$$
$$\mathbf{D} = \mathbf{B_2C_1} + \mathbf{D_1D_2}$$

$$T_1 = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \qquad T_2 = \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} \qquad T = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$T = T_1 T_2$$